

Novel primers for sequencing of the complete mitochondrial cytochrome *b* gene of ungulates using non-invasive and degraded biological samples

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Abstract We describe a set of novel primers for successful amplification of the complete mitochondrial cytochrome *b* (cyt *b*) gene of ungulate species. DNA extracted from non-invasively obtained and decomposed samples is found to be degraded and inappropriate for amplification of the complete gene (more than 1 kb) in single PCR amplification. We developed a series of six ungulate-specific conserved primers for the cyt *b* gene. These primers, in various combinations, amplify 366–1,266 bp fragments. We also used them in multiplex PCR reaction to assess the maximum possible size of the PCR amplification product.

Keywords Mitochondrial cytochrome *b* gene · Non-invasive and decomposed samples · PCR amplification · Ungulate species · Multiplex PCR

Introduction

DNA sequencing-based species identification relies on the amount and veracity of the sequence data available at the appropriate database. Identification of species from a sample of unknown origin is based on comparison of DNA sequences obtained from the sample against existing sequences in databases (Verma and Singh 2003; Gupta et al. 2005). A lack of reference sequences from the matching species in the database may lead to high-scoring matches with closely related sequences and produce unreliable results (Naidu et al. 2012). The International Society of Forensic Genetics (ISFG) recommends the use of an in-

house and authentic DNA database for species identification for forensic validation (Linacre et al. 2011). Mitochondrial DNA (mtDNA) cytochrome *b* (cyt *b*) is an appropriate molecular marker for species identification (Tobe et al. 2010). Hence, large amounts of reference sequence data for the complete cyt *b* gene are strongly needed for phylogenetic and forensic studies. The use of a conserved primer will be extremely useful for generating a database.

The mitochondrial genome is a widely preferred locus for the development of conserved primers (Kocher et al. 1989; Verma and Singh 2003). Such primers have proved to be valuable for species identification in the wildlife forensics (Gupta et al. 2005). Recently, a pair of conserved primer for the complete cyt *b* gene was developed for mammals (Naidu et al. 2012). Amplifying the complete cyt *b* gene requires high-quality DNA; however, conservation and forensic geneticists often depend upon non-invasively obtained and forensic (mostly degraded) biological samples. Amplification and analysis of longer fragments (more than 1 kb) from DNA extracted from such samples is challenging. A primer amplifying shorter fragments will be more useful for such studies. We are working on the genetics of ungulates. Hence, we describe novel primers for amplification of the complete mtDNA cyt *b* gene of these animals from degraded samples.

Complete cyt *b* sequences of various ungulate species were obtained from GenBank and aligned using ClustalW (Thompson et al. 1994). Three forward and three reverse primers were designed on the basis of sequence homology (Table 1) and procured (Sigma Life Sciences, India).

Biological samples received for species identification in cases of ungulate poaching and fecal pellets of deer species were used. DNA was extracted (Gupta et al. 2013) and

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Table 1 Six primers designed for amplification of either complete or partial fragment of mtDNA cyt *b* gene

S. no.	Primer name	Sequence (5' → 3')
1	MC <i>b</i> (14081-105) F1	CATTATTCTCACATGGAATCTAACC
2	MC <i>b</i> (14560-584) F2	GAGGACAAATATCATTTCTGAGGAGC
3	MC <i>b</i> (14584-560) R1	GCTCCTCAGAATGATATTTGTCCTC
4	MC <i>b</i> (14981-934) F3	ACCCAGACAACCTACACCCAGCAA
5	MC <i>b</i> (14934-981) R2	TTACTGGGGTGTAGTTGTCTGGGT
6	MC <i>b</i> (15347-323) R3	CTCCTTTTCTGGTTTACAAGACCAG

Primer position is derived from the complete mtDNA sequence of *Rusa unicolor swinhoi* (EF035448)

Table 2 All possible combinations of the primers during PCR amplification and expected amplicon lengths

S. no.	Primer combination	Expected amplicon length(s) (bp)
1	MC <i>b</i> (14081-105) F1 and MC <i>b</i> (14584-560) R1	503
2	MC <i>b</i> (14081-105) F1 and MC <i>b</i> (14934-981) R2	853
3	MC <i>b</i> (14081-105) F1 and MC <i>b</i> (15347-323) R3	1,266
4	MC <i>b</i> (14560-584) F2 and MC <i>b</i> (14934-981) R2	374
5	MC <i>b</i> (14560-584) F2 and MC <i>b</i> (15347-323) R3	787
6	MC <i>b</i> (14981-934) F3 and MC <i>b</i> (15347-323) R3	366
<i>In multiplex reactions</i>		
7	F1, R1, R2 and R3	503, 853 and 1,266
8	F2, R2 and R3	374 and 887
9	F1, F2, R2 and R3	374, 787, 853 and 1,266
10	F1, F2, F3 and R3	366, 787 and 1,266

Table 3 List of tested species for validation of primer

S. no.	Amplification and sequencing with ungulate species	Result	No. of individuals
1	Sambar (<i>Rusa unicolor</i>)	Successful	9
2	Swamp deer (<i>Rucervus duvaucelii</i>)	Successful	2
3	Chital (<i>Axis axis</i>)	Successful	3
4	Hog deer (<i>Axis porcinus</i>)	Successful	3
5	Barking deer (<i>Muntiacus muntjak</i>)	Successful	2
6	Black buck (<i>Antelope cervicapra</i>)	Successful	2
7	Indian gazelle (<i>Gazella bennettii</i>)	Successful	2
8	Nilgai (<i>Boselaphus tragocamelus</i>)	Successful	4
9	Indian wild pig (<i>Sus scrofa</i>)	Successful	8
10	Domestic pig (<i>Sus scrofa</i>)	Successful	6
<i>Amplification with non ungulate species</i>			
11	Tiger (<i>Panthera tigris</i>)	No amplification	
12	Leopard (<i>Panthera pardus</i>)	No amplification	
13	Human (<i>Homo sapiens</i>)	No amplification	
14	Common langur (<i>Semnopithecus entellus</i>)	No amplification	
15	Palm civet (<i>Viverricula indica</i>)	No amplification	
16	Different fish species	No amplification	

subjected to PCR amplification using the primer combinations listed in Table 2. The amplification was carried out in a 20 µl reaction volume containing 1 µl of the extracted DNA, 100 µM of dNTPs, 4 pmol of each primer, 1.5 mM MgCl₂, 0.5 units of AmpliTaq Gold (Invitrogen Life Technologies), and 1 × PCR buffer (10 mM Tris–HCl, pH

8.3, and 50 mM KCl). The PCR conditions were the following: initial denaturation at 95 °C for 10 min, followed by 35 cycles of denaturation at 95 °C for 45 s, annealing at 55 °C for 1 min, and extension at 72 °C for 1.5 min. The final extension was at 72 °C for 10 min. The PCR products were electrophoresed on 2 % agarose gel, stained with

ethidium bromide (0.5 mg/ml), and visualized under a UV transilluminator.

The PCR products obtained were sequenced (ABI 3130, Genetic Analyser) on both strands. The sequences were aligned using SeqScape v2.5 (Applied Biosystems) and blasted using the program BLAST to confirm the origin of the sample (Altschul et al. 1997).

Results and conclusion

PCR amplicons of expected length were obtained with different primer combinations (Table 2). Shorter fragments of length 366, 374, and 503 bp could be amplified with the DNA from all the biological samples, including fecal pellets (Table 2). A PCR product of length 1,266 bp was amplified with the DNA extracted from fresh and less decomposed samples of flesh. However, in the majority of moderately putrefied flesh samples, amplicons of length 853 and 787 bp were obtained. After alignment of the DNA sequences obtained from all the above amplicons, complete gene sequences were obtained for various ungulate species. A complete list of the species tested and the corresponding results are provided in Table 3.

Enrichment of DNA sequences in the database of forensic laboratories and GenBank is one of the essential steps for the success of forensic work and identification of species with precision. For successful comparison of sequences of unknown biological samples, either suitable fragments or complete *cyt b* gene sequences from various animal species are essential. Obtaining permissions for collecting biological samples invasively from free-ranging animals of endangered and protected species is a challenging task. However, collection of non-invasive samples is an efficient way of increasing the sample size. Hence, amplification of smaller fragments and alignment of sequences is a convenient way of enhancing databases for species for which samples are difficult to obtain. The availability of novel and complete gene sequences can

improve the reliability of sequence-based species identification. The primers described here will be beneficial for the amplification and sequencing of the complete *cyt b* gene of various ungulate species using non-invasively obtained biological samples.

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